DBMS Assignment 1-4
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Section-A

1 What is transaction in DBMS?

Ans A transaction is a sequence of database operation performed as a single logical unit of work. It must be completed fully or not all to maintain data correctness.

Ex:- Money transfer between bank accounts.

2. Explain ACID properties.

Ans: ACID ensure reliable transaction

- Atomicity: All operation complete or none.
- Consistency: Database stays correct before & after transaction
- Isolation: Concurrent transactions don't interfere.
- Durability: Committed changes are permanent.

Example: Two users booking tickets. Online payment system.

3 What is concurrency control?

Ans:- Concurrency control manage multiple transaction at the same time without conflicts. It prevent issue like lost updates, dirty reads, and inconsistency.

4. What is a deadlock?

Ans Deadlock occurs when two or more transactions wait for each other forever, so none can proceed.

Example: T_1 holds lock A & waits for B, while T_2 holds B & wait for A

5. What is serializability? 6. What is recovery in DBMS?

Ans Recovery is the process of restoring the database to a correct state after a crash, power failure, or error. It uses logs, checkpoints undo, & redo operations.

→ Serializability mean the result of concurrent transaction execution should be the same as if the transaction were executed one after another. It ensure correctness.

Section-B

1 Explain ACID properties using banking example?

Ans Suppose 1000 is transfered from account A to account B.

- Atomicity :- Either ₹1000 is deducted from A & added to B, or no change happen
- Consistency :- Total money in both account remains same
- Isolation :- If two transfer happen together, they should not affect each other
- Durability :- Once transfer is successful, it is permanently stored.

This ensure safe & reliable banking transaction.

2. Illustrate a deadlock situation with two transaction.

Ans Example

- T_1 locks resource A & want resource B.
- T_2 locks resource B & want resource A.

Now.

- T_1 waits for B.
- T_2 waits for A.

Both waits forever → Deadlock occurs.

Solutions :- Deadlock detection, timeout or prevention methods.

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3. Explain serial vs concurrent execution

- Serial Execution: Transactions run one after another
  - Safe but slow.

- Concurrent Execution: Transactions run together.
  - faster but may cause conflicts.

Ex: Serial: T_1 finishes, then T_2 starts
Concurrent: T_1 & T_2 run at same time

Section: C

1. Banking system transfer & ACID.

Ans: In bank transfer:
- Debit sender account
- Credit receiver account

ACID role:
Atomicity: both steps complete together.
Consistency: balance remains correct.
Isolation: no interference from other transaction.
Durability: saved permanently.
Thus, transaction remains safe.

2. Two transaction update can same data simultaneously.

Ans: Possible issue:-
Lost Update: One update overwrite another.
Dirty Read: Reading uncommited data
Inconsistent: Wrong final result.

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Solution:- Uses locks, timestamps & isolation levels

Q3. System crashes during transaction recovery steps.
Recovery steps:
1 Check transaction log.
2 Undo incomplete transaction (Roll back).
3 Redo complete transaction if needed.
4 Restore from checkpoint.
The brings database back to consistent state.

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